

Ex.no:5b	DATA VISUALIZATION USING MATPLOTLIB – SALES DATA

AIM

To visualize and analyze the **California Housing Dataset** using Line Chart, Bar Chart, and Pie Chart with Matplotlib to understand housing value trends, ocean-proximity-wise comparisons, and housing distribution across different locations.

ALGORITHM

Step 1: Import Pandas and Matplotlib libraries.

Step 2: Load the California Housing Dataset into a DataFrame.

Step 3: Group the housing data according to ocean proximity and calculate the average median house value for each category.

Step 4: Plot the ocean-proximity-wise house values using a Line Chart to observe variations in house values across different locations.

Step 5: Group the dataset according to ocean proximity and calculate the total number of housing records for each category.

Step 6: Plot the housing records using a Bar Chart to compare the number of houses across different ocean proximity categories.

Step 7: Select the major ocean proximity categories and calculate their proportions in the dataset.

Step 8: Plot the housing distribution using a Pie Chart.

Step 9: Add suitable titles, axis labels, legends, and display the visualizations.

CODE

```
import pandas as pd
import matplotlib.pyplot as plt

# Load California Housing Dataset
df = pd.read_csv(
```

```
"housing prices.csv",
low_memory=False
)

# -----
# 1. Line Chart – House Value by Ocean Proximity
# -----

house_value = (
    df.groupby("ocean_proximity")["median_house_value"]
    .mean()
    .sort_values(ascending=False)
)

plt.figure(figsize=(10, 5))

plt.plot(
    house_value.index,
    house_value.values,
    marker="o"
)

plt.title("Average House Value by Ocean Proximity")
plt.xlabel("Ocean Proximity")
plt.ylabel("Average Median House Value")
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()

plt.show()
```

```
# -----  
  
# 2. Bar Chart – Houses by Ocean Proximity  
# -----  
  
house_count = (  
    df["ocean_proximity"]  
    .value_counts()  
    .sort_values(ascending=False)  
)  
  
plt.figure(figsize=(12, 6))  
  
plt.bar(  
    house_count.index,  
    house_count.values  
)  
  
plt.title("Number of Houses by Ocean Proximity")  
plt.xlabel("Ocean Proximity")  
plt.ylabel("Number of Houses")  
plt.xticks(rotation=45, ha="right")  
plt.tight_layout()  
  
plt.show()  
  
# -----  
  
# 3. Pie Chart – Housing Distribution  
# -----  
  
housing_distribution = (  
    df["ocean_proximity"]
```

```

        .value_counts()
        .sort_values(ascending=False)
    )

plt.figure(figsize=(8, 8))

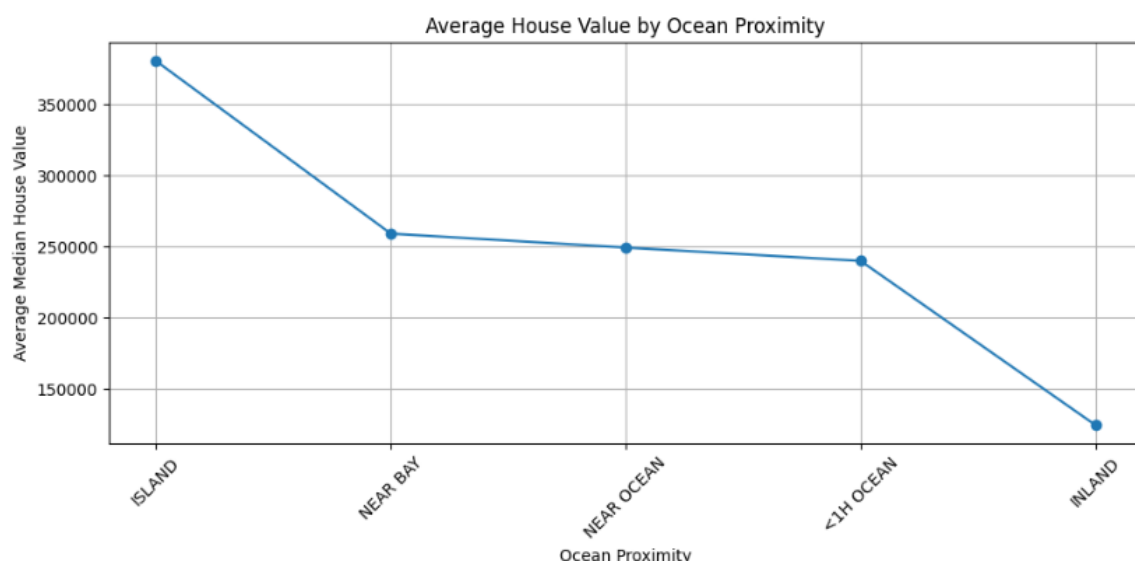
plt.pie(
    housing_distribution.values,
    labels=housing_distribution.index,
    autopct="%1.1f%%"
)

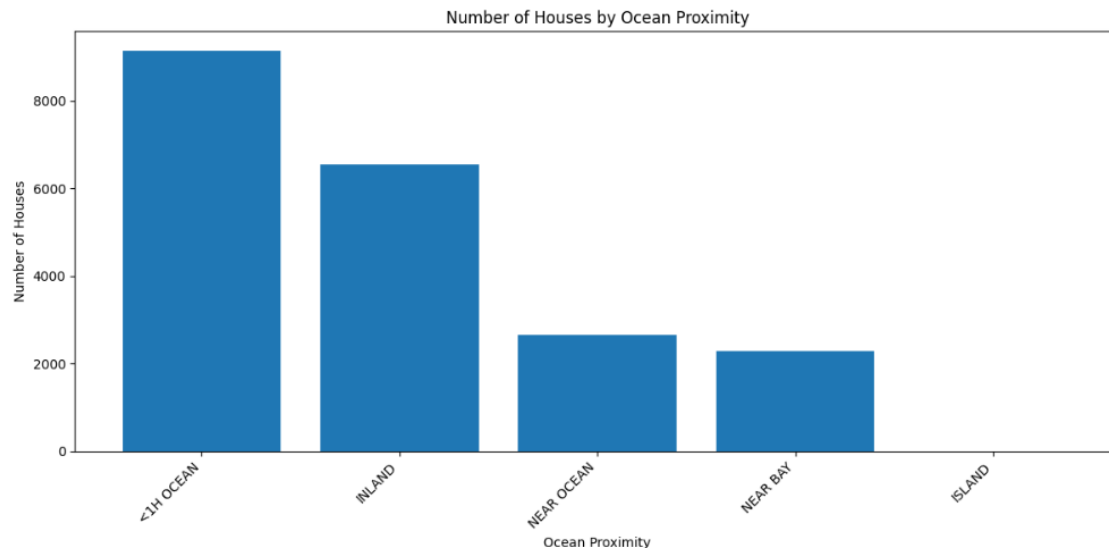
plt.title("Housing Distribution by Ocean Proximity")

plt.show()

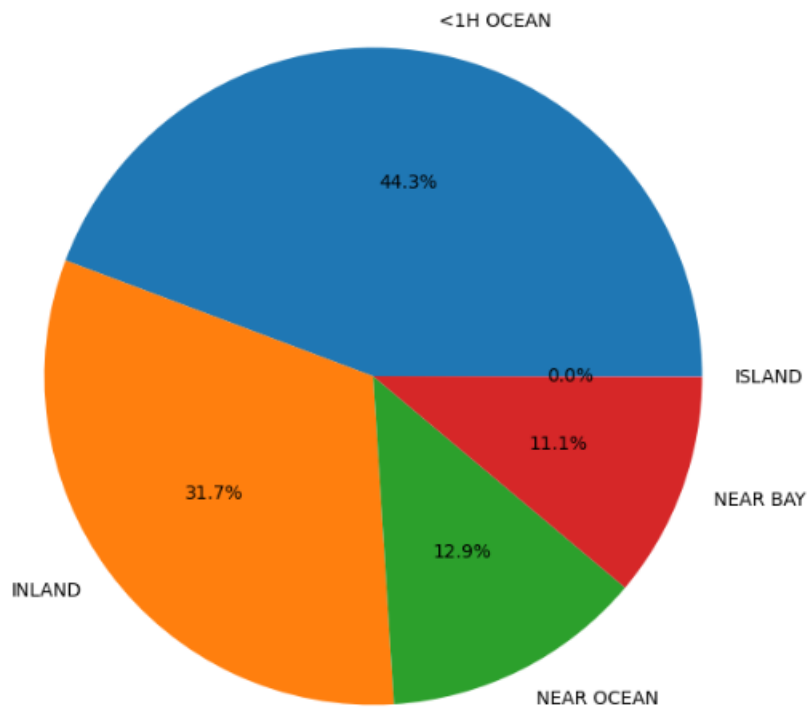
```

OUTPUT





Housing Distribution by Ocean Proximity



RESULT